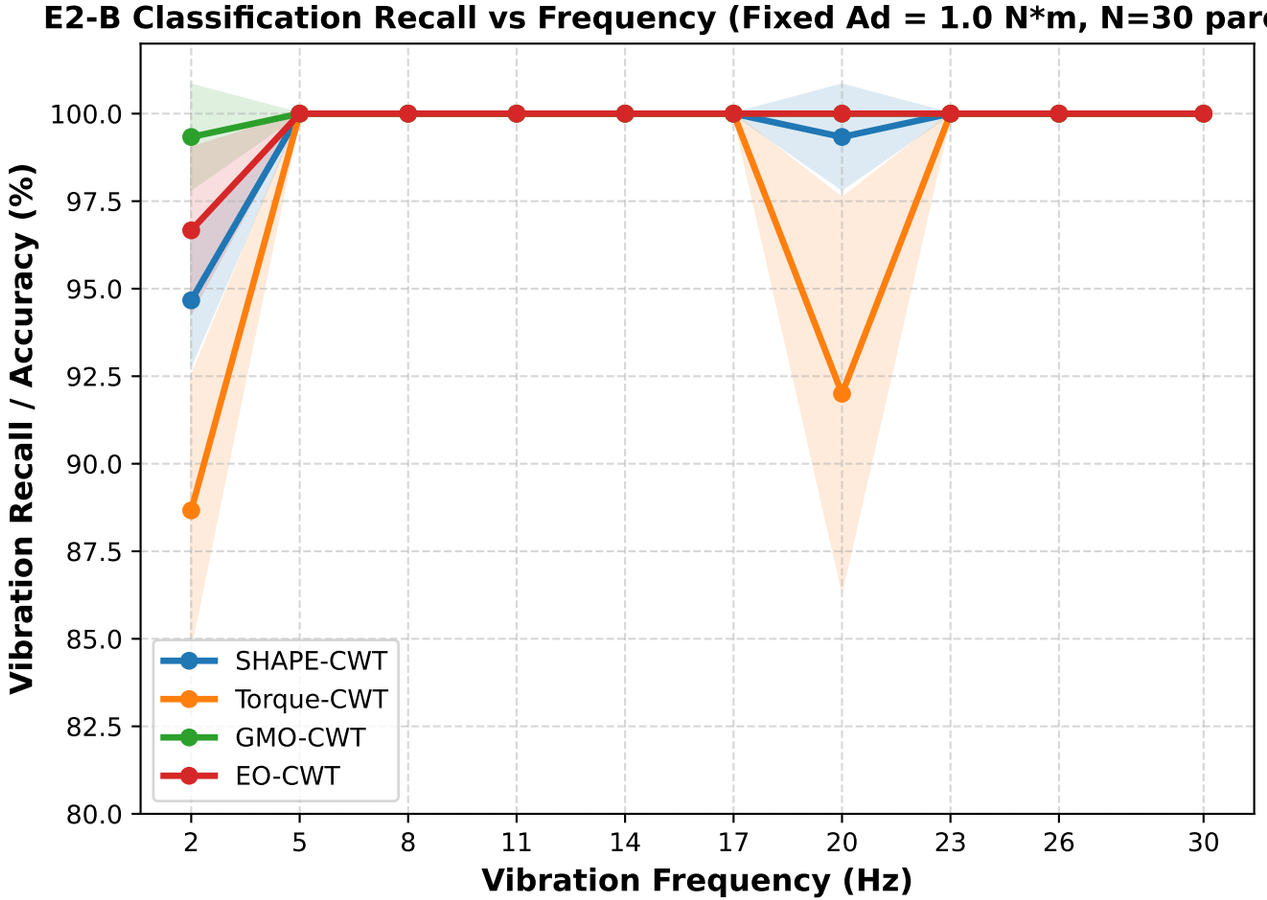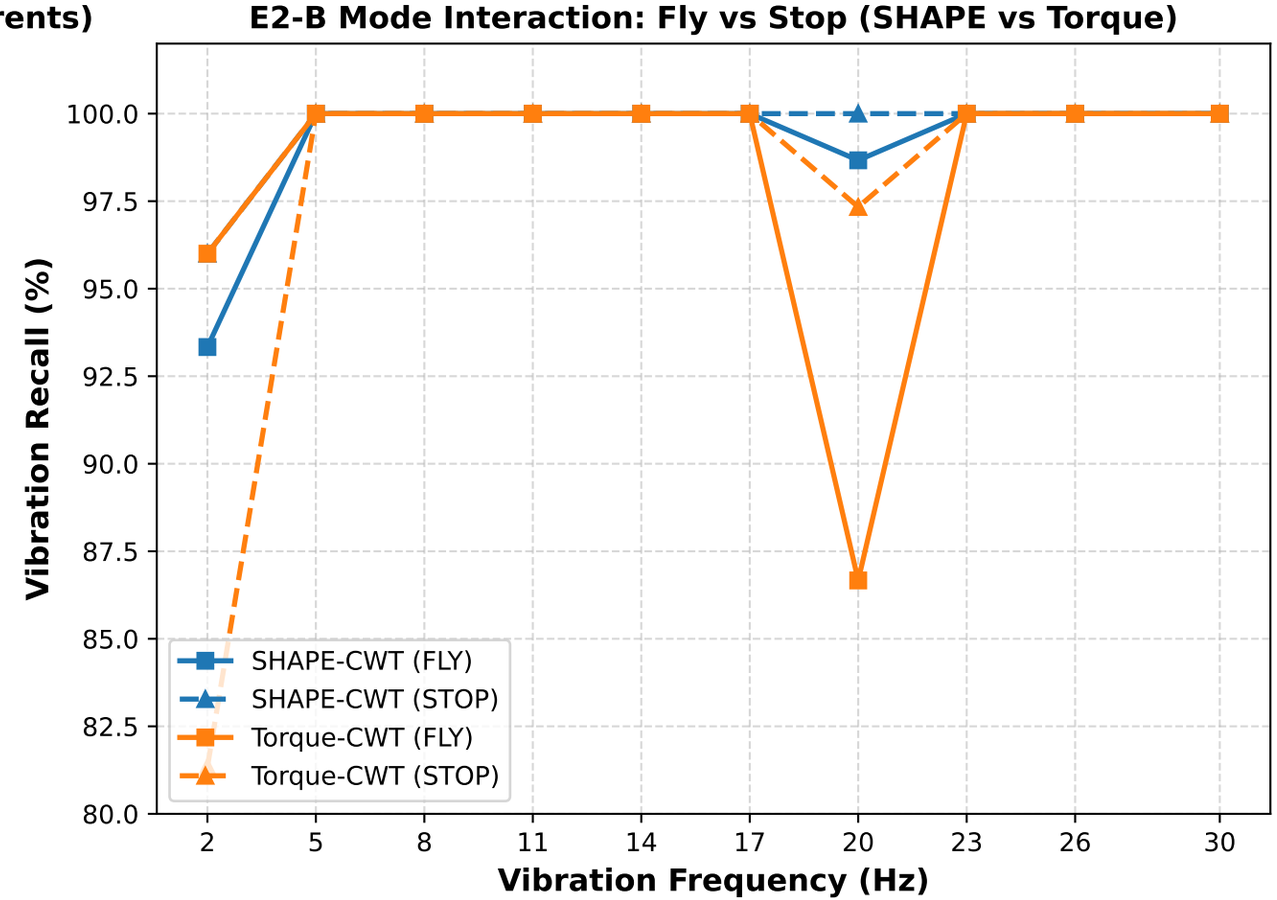E2-B Controlled Frequency Sweep Analysis